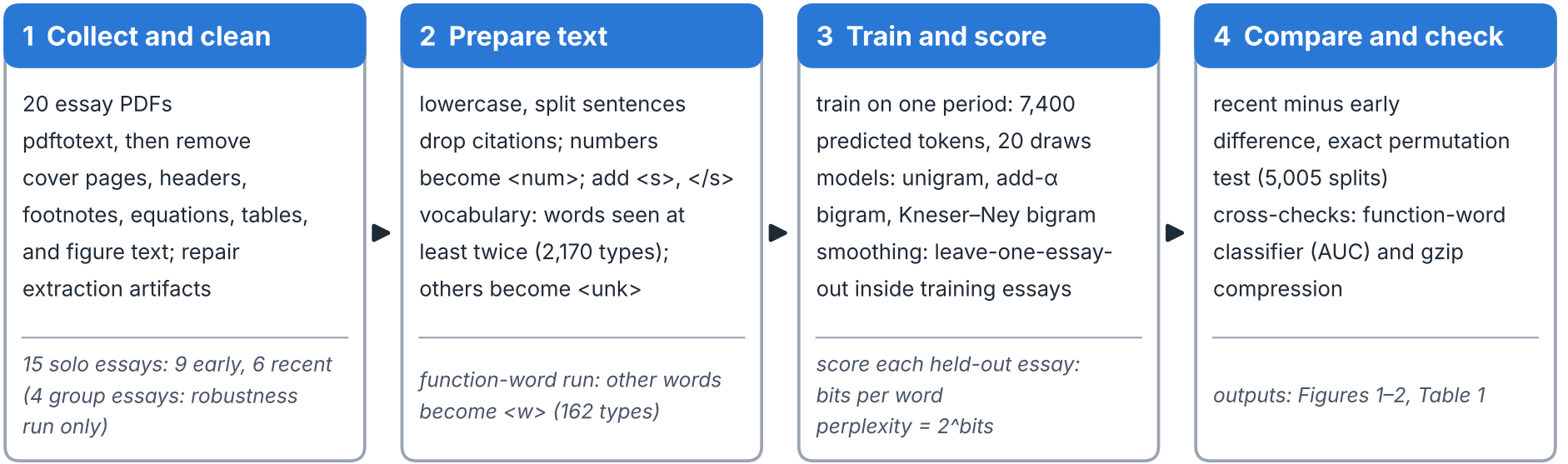
1 Collect and clean
20 essay PDFs
pdftotext, then remove cover pages, headers, footnotes, equations, tables, and figure text; repair extraction artifacts
15 solo essays: 9 early, 6 recent (4 group essays: robustness run only)
2 Prepare text
lowercase, split sentences
drop citations; numbers become <num>; add <s>, </s>
vocabulary: words seen at least twice (2,170 types); others become <unk>
function-word run: other words become <w> (162 types)
3 Train and score
train on one period: 7,400 predicted tokens, 20 draws
models: unigram, add-α bigram, Kneser–Ney bigram
smoothing: leave-one-essay-out inside training essays
score each held-out essay:
bits per word
perplexity = 2^bits
4 Compare and check
recent minus early difference, exact permutation test (5,005 splits)
cross-checks: function-word classifier (AUC) and gzip compression
outputs: Figures 1–2, Table 1